C22 Period 3 Elements and Oxides – Knowledge Organiser

AQA A-level Chemistry • Set 1: Reactions of the Elements **Paper 1**

#	Question	Answer
1	Equation for Na with water	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
2	Equation for Mg with cold water	$Mg + 2H_2O \rightarrow Mg(OH)_2 + H_2$ (very slow)
3	Equation for Na with oxygen	$4Na + O_2 \rightarrow 2Na_2O$
4	Flame colour when Na burns	Yellow
5	Flame colour when Mg burns	Bright white
6	Flame colour when S burns	Blue
7	Product of P burning in excess oxygen	P_4O_{10}
8	Product of S burning in oxygen	SO_2
9	Formula of aluminium oxide	Al_2O_3
10	Formula of silicon dioxide	SiO_2

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AQA A-level Chemistry • Set 2: Structure and Type of Oxides **Paper 1**

#	Question	Answer
1	Structure of Na_2O, MgO and Al_2O_3	Giant ionic
2	Structure of SiO_2	Giant covalent
3	Structure of P_4O_{10}, SO_2 and SO_3	Simple molecular
4	Period 3 oxide with highest melting point	MgO
5	Why MgO melts higher than Na_2O	Mg^{2+} has higher charge
6	Basic oxides	Na_2O, MgO
7	Amphoteric oxide	Al_2O_3
8	Acidic oxides	SiO_2, P_4O_{10}, SO_2, SO_3
9	Amphoteric	Reacts with acids and bases
10	Period 3 oxides insoluble in water	Al_2O_3, SiO_2

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AQA A-level Chemistry • Set 3: Oxide Equations **Paper 1**

#	Question	Answer
1	Na_2O with water (and pH)	$Na_2O + H_2O \rightarrow 2NaOH$; pH 14
2	MgO with water (and pH)	$MgO + H_2O \rightarrow Mg(OH)_2$; pH 9
3	P_4O_{10} with water (and pH)	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$; pH 0
4	SO_2 with water (and pH)	$SO_2 + H_2O \rightarrow H_2SO_3$; pH 2
5	SO_3 with water (and pH)	$SO_3 + H_2O \rightarrow H_2SO_4$; pH 0
6	Al_2O_3 with HCl	$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$
7	Al_2O_3 with NaOH	$Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4$
8	SiO_2 with NaOH	$SiO_2 + 2NaOH \rightarrow Na_2SiO_3 + H_2O$
9	MgO with HCl	$MgO + 2HCl \rightarrow MgCl_2 + H_2O$
10	SO_2 with NaOH	$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$